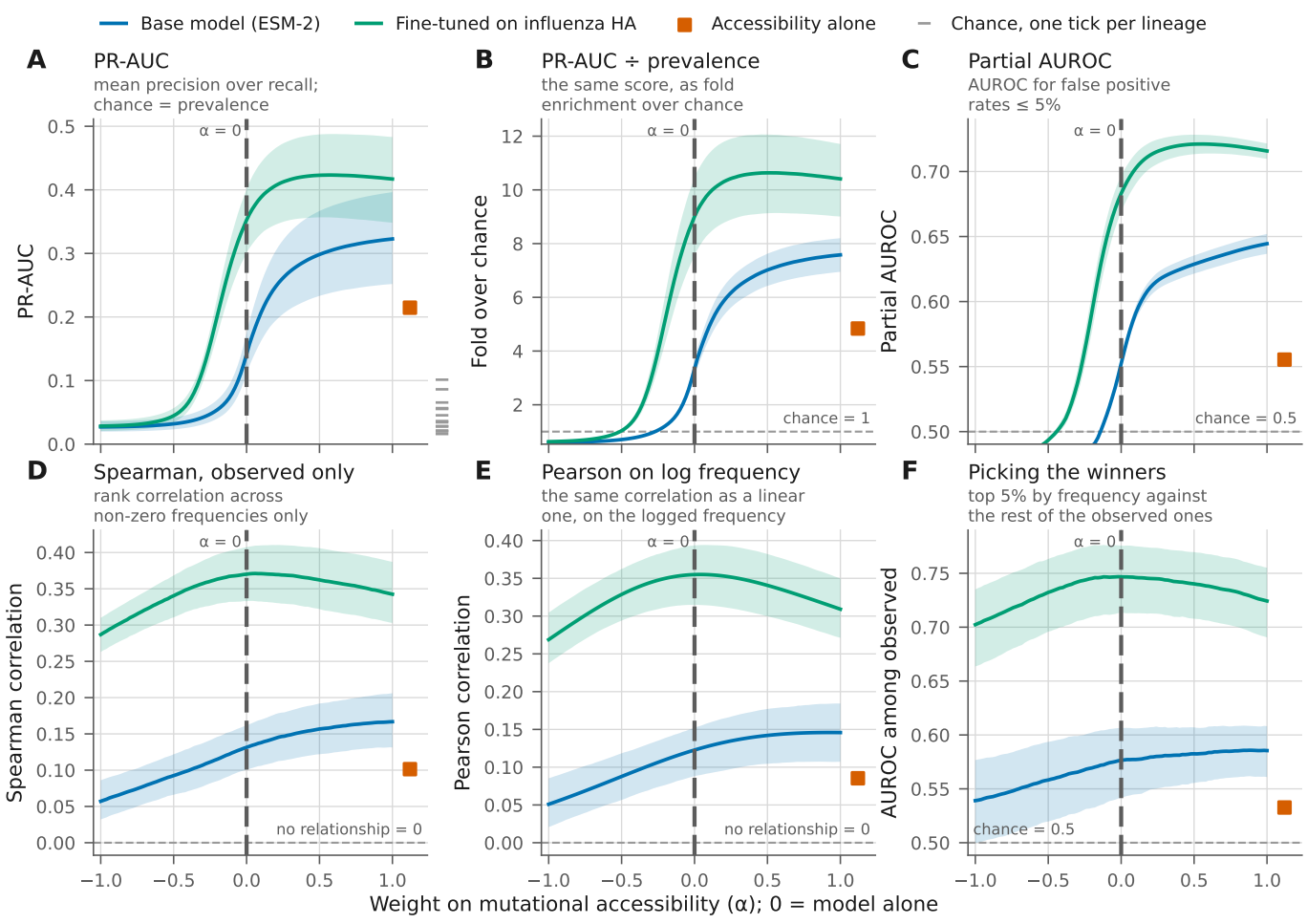

Base model (ESM-2)
Fine-tuned on influenza HA
Accessibility alone
Chance, one tick per lineage
A
PR-AUC
mean precision over recall;
chance = prevalence
PR-AUC
α = 0
B
PR-AUC ÷ prevalence
the same score, as fold
enrichment over chance
Fold over chance
chance = 1
C
Partial AUROC
AUROC for false positive
rates ≤ 5%
Partial AUROC
chance = 0.5
D
Spearman, observed only
rank correlation across
non-zero frequencies only
Spearman correlation
no relationship = 0
E
Pearson on log frequency
the same correlation as a linear
one, on the logged frequency
Pearson correlation
F
Picking the winners
top 5% by frequency against
the rest of the observed ones
AUROC among observed
Weight on mutational accessibility (α); 0 = model alone